MATH 111 Calculus 1

The limit of a function

Week 1 · Lecture 1 · Stewart §1.5

Dr. Salnikova Chekeres · New Uzbekistan University · Fall 2026

While you wait

Please fill the front rows and sit next to three other people. Those three are your team for the whole semester. Have your notebook open.

STAND UP

Stand up if you cannot hear me well.

(Yes — this instruction is for the people who cannot hear me. That is why it is on the screen.)

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Everyone standing: please move to the front rows now.

The back third of this hall is closed today. Thank you.

How this hall works

- This hall is bad for voices. So we will talk with hands, feet and one microphone.
- Every question I ask, all 200 of you answer — with your hand, by standing up, or on a page of your notebook.
- You work in a team of four that stays together all semester.
- Every week one lecture question reappears, word for word, on the weekly quiz.
- Some days this will feel harder than listening. That feeling is what learning feels like.

The five commands



THINK	The clock runs. Silence. Write your letter (A, B, C or D) in your notebook.
VOTE	Eyes on the screen. Hands up when your letter appears; hold three seconds.
TALK	Show your team your answer with fingers (1–4). Convince each other. Two minutes.
VOTE again	Same as before. Watch how the hall changes.
MIC	A team gets the microphone and gives the reason. I repeat every answer into my microphone.

Two more: **STAND UP** if a statement is true; **EXIT TICKET** — a quarter page from your notebook at the end.

PRACTICE VOTE

Honestly: the biggest obstacle to learning in this hall is …

A. the acoustics B. the temperature C. hunger D. the person sitting next to me

(Nobody is counting. Well — I am, roughly. That is the point.)

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Hands down. Thank you.



Your team

Now (2 minutes)

- Whatever you just voted: the person next to you is now on your team. Four people, all semester.
- Tear one page from a notebook: team name, four names (with your group). A runner collects it at the end.
- Team numbers arrive next week; this week the microphone finds you by row.

Roles (rotate every week)

- **Courier** — speaks into the microphone
- **Scribe** — writes the exit ticket
- **Sceptic** — asks “why?”
- **Timekeeper** — watches the clock

Your team

Now (2 minutes)

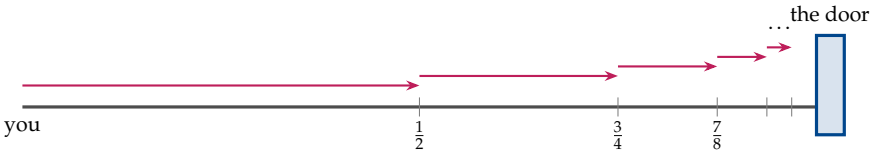
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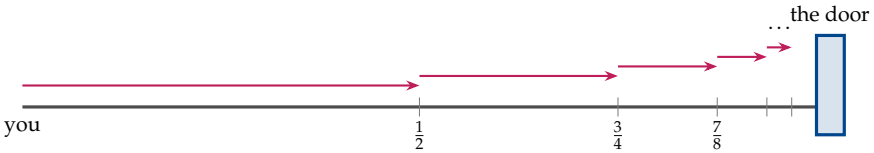
I need two volunteers to be today's runners (you carry the microphone).
Hands up!

Why we need a new idea: the door problem



To leave this hall you must first walk half the way to the door.

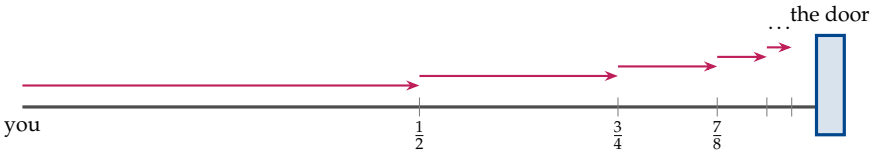
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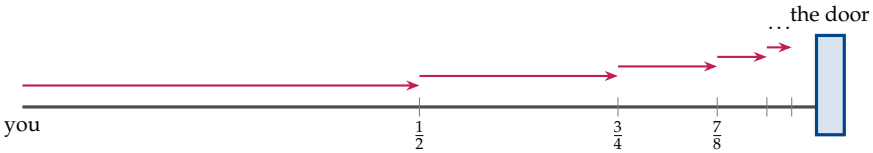


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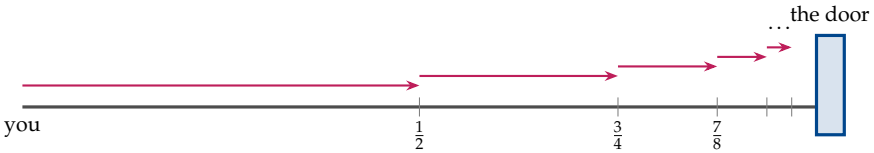
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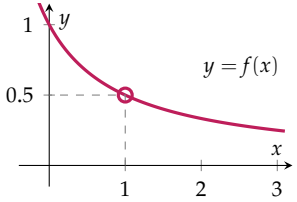
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So nobody can ever reach the door? (Zeno, about 450 BC, was quite pleased with this.)

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Calculus is the art of finishing infinitely many steps in finite time. The tool that makes it honest is today's topic: the limit.

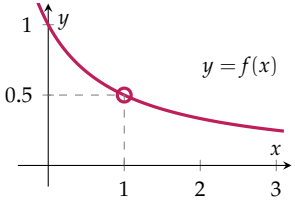
Our first limit: a graph and a table



$$f(x) = \frac{x-1}{x^2-1}$$

What does $f(x)$ do as x approaches 1? (At $x = 1$ itself: $0/0$, not defined — the hole.)

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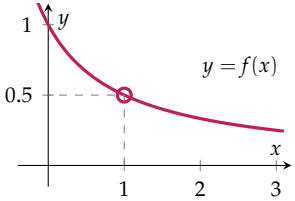


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0.99	0.502513	1.01	0.497512
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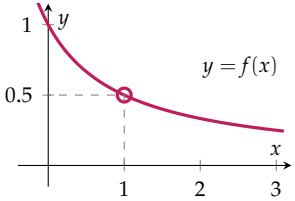
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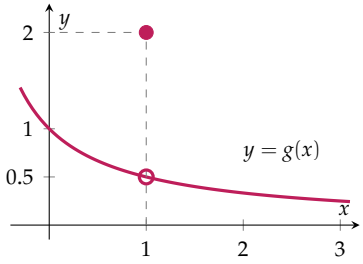
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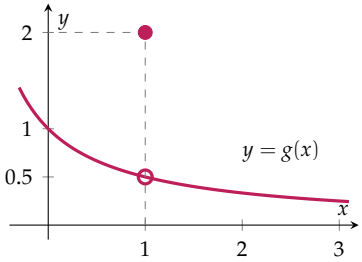
We write $\lim_{x\to 1} \frac{x-1}{x^2-1} = \frac{1}{2}$: *the limit of* $f(x)$*, as* x *approaches* 1*, is* $\frac{1}{2}$.

Change the function at one point — what changes?



$$g(x) = \begin{cases} \dfrac{x-1}{x^2-1}, & x \neq 1, \\ 2, & x = 1. \end{cases}$$

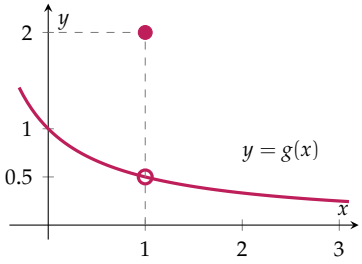
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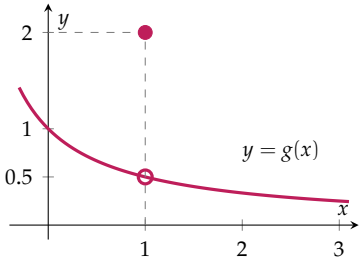


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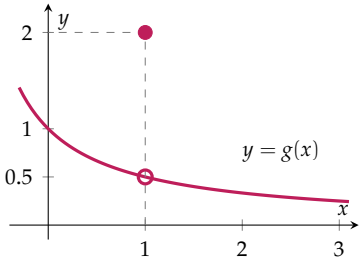
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Keep this in mind for the first question.

A little history · 1786 and 1908

The word "lim" was first written by Simon L'Huilier in 1786 — with a full stop, "lim.", as an abbreviation. The arrow in $x \to a$ is much younger: it appeared in English textbooks in 1905–1908, and G. H. Hardy's *Course of Pure Mathematics* made it standard. For a century people wrote the limit with "$x = a$" underneath and had to explain that they did not mean $x = a$.

Question 1

Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2, \\ 7, & x = 2. \end{cases}$ and some values near 2:

$f(1.9) = 3.9,\ \ f(1.99) = 3.99,\ \ f(2.01) = 4.01,\ \ f(2.1) = 4.1.$

$$\lim_{x \to 2} f(x) = \ ?$$

A. 4

B. 7

C. It does not exist, because $f(2) \neq 4$.

D. 0

THINK · silence · write your letter

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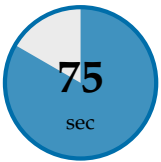
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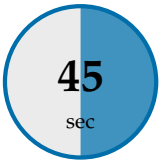
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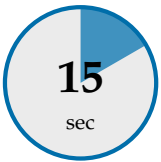
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120 sec

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Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

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Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

Question 1

TALK · 2 minutes

Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2, \\ 7, & x = 2. \end{cases}$ and some values near 2:

$f(1.9) = 3.9$, $f(1.99) = 3.99$, $f(2.01) = 4.01$, $f(2.1) = 4.1$.

$\lim_{x\to 2} f(x) = \ ?$

A. 4 **B.** 7

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A. 4

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D. 0



In your team

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VOTE AGAIN

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A. 4 **B.** 7

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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

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A B C **D**

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Hands down. Thank you.

A B C D

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The microphone goes to

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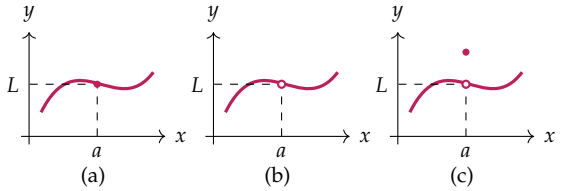
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Answer

A. For $x \neq 2$: $f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2 \to 4$. The value $f(2) = 7$ is irrelevant — a limit looks *near* 2, never *at* 2. (B: value instead of limit. C: believes the limit must equal the value.)

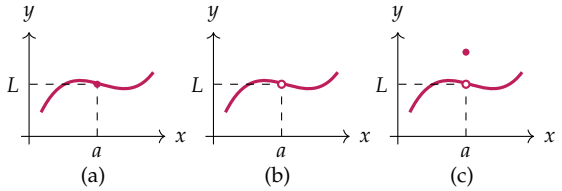
Definition (intuitive)



Limit of a function

We write $\lim_{x \to a} f(x) = L$ if we can make $f(x)$ as close to L as we like by taking x close enough to a (on either side), but $x \neq a$.

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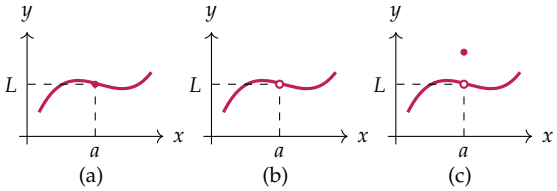


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A limit is about the approach, not the arrival — remember Zeno.

Guessing from a table — a guess that works

$$\lim_{x \to 0} \frac{\sin x}{x} = \; ? \qquad (x \text{ in radians})$$

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± 0.5	0.95885
± 0.1	0.99833
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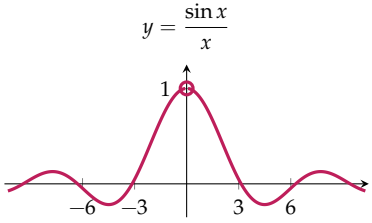
The table suggests $\lim_{x \to 0} \frac{\sin x}{x} = 1$ — and that is correct (we prove it in a few weeks).

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$x = \frac{2}{4n+1}$ gives 1 every time,
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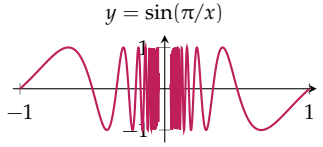
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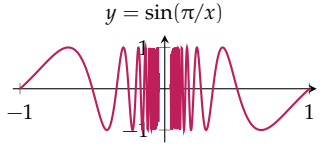
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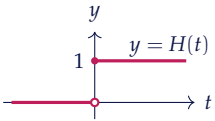
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Tables suggest; they never prove.

Definitions

Left-hand limit: $\lim_{x \to a^-} f(x) = L$ means $f(x) \to L$ as $x \to a$ with $x < a$.

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One-sided limits

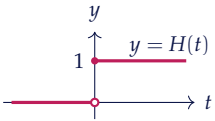
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$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^-} f(x) = L$ and $\lim_{x \to a^+} f(x) = L$.



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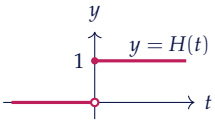
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A light switch flipped at $t = 0$: $H(t) = 0$ for $t < 0$, $H(t) = 1$ for $t \geq 0$.



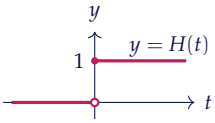
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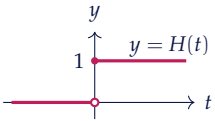
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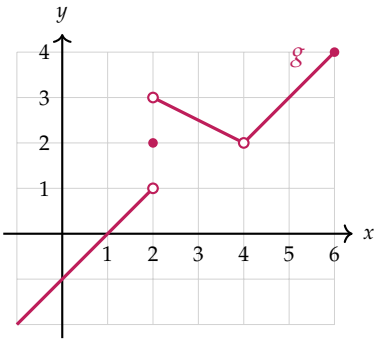
$\lim_{t \to 0^-} H(t) = 0$ but $\lim_{t \to 0^+} H(t) = 1$: different sides, different answers.

So $\lim_{t \to 0} H(t)$ does not exist — although $H(0) = 1$ is defined.

A little history · 1890s

Oliver Heaviside was a self-taught telegraph engineer who never went to university; he invented this function to describe switching a current on. Academic mathematicians complained that his methods were not rigorous. His answer: "Shall I refuse my dinner because I do not fully understand the process of digestion?"

Question 2



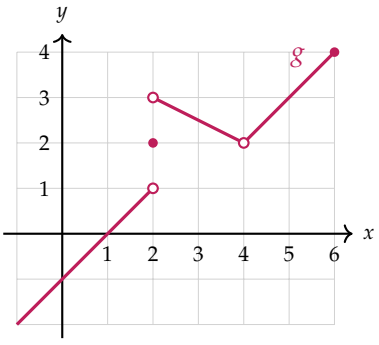
The graph of g is shown.

$$\lim_{x \to 2} g(x) = \ ?$$

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C. It does not exist. **D.** 2

THINK · silence · write your letter



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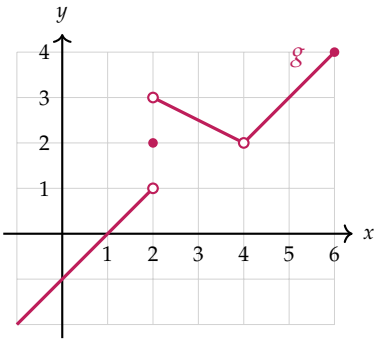
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Question 2

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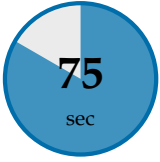


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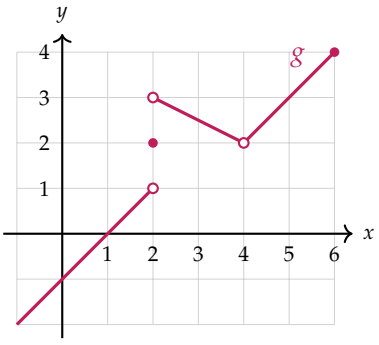
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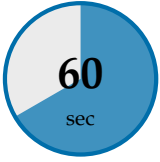


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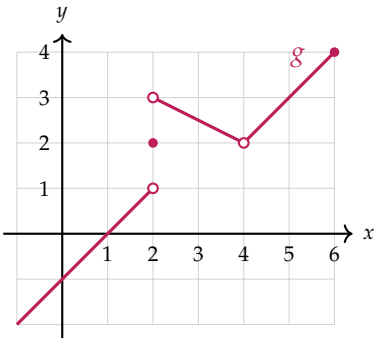
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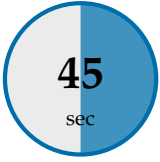


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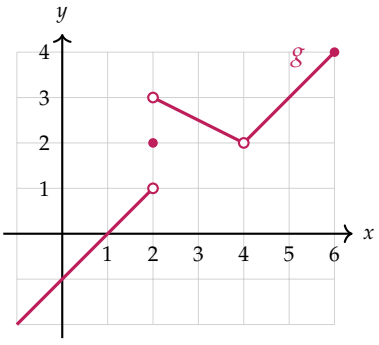
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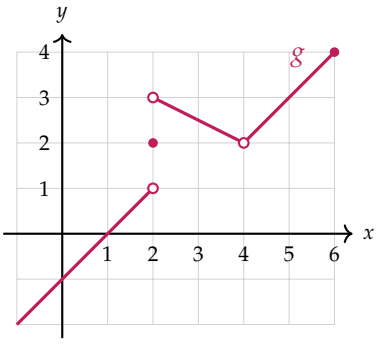
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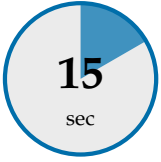


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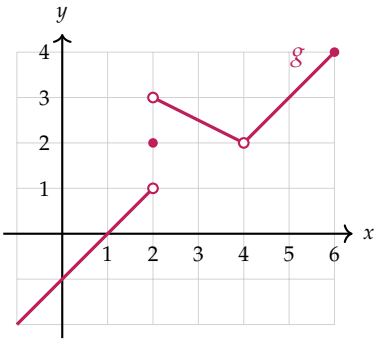
$$\lim_{x \to 2} g(x) = \ ?$$

A. 1 **B.** 3

C. It does not exist. **D.** 2



THINK · silence · write your letter



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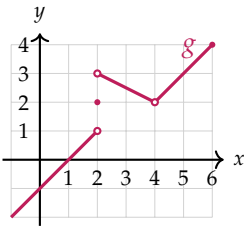
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C. It does not exist. **D.** 2

0
sec

VOTE



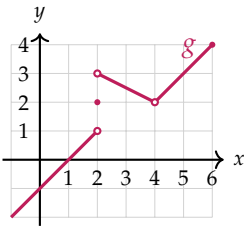
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE



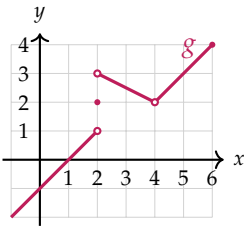
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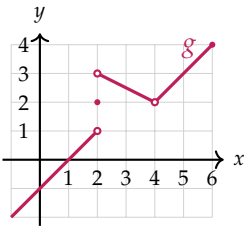
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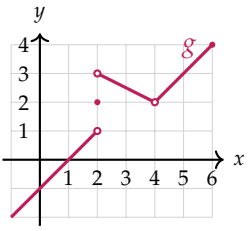
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A B **C** D

VOTE



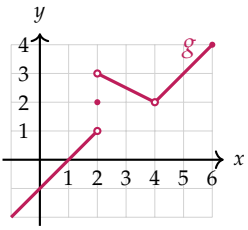
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A B C **D**

VOTE



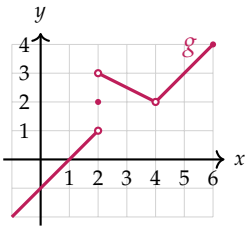
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Hands down. Thank you.

A B C D

TALK · 2 minutes



The graph of g is shown. $\lim_{x \to 2} g(x) = \ ?$

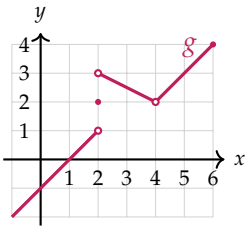
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120 sec

In your team

Fingers up: 1 = A, 2 = B, 3 = C, 4 = D. Disagree? Convince each other. Agree? Find the reason together. Sceptic: ask "why?".

TALK · 2 minutes



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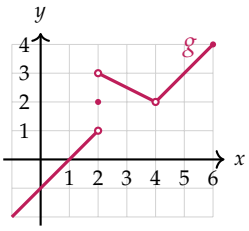
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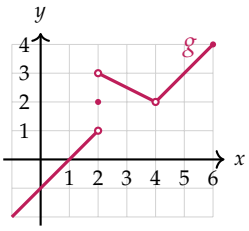
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TALK · 2 minutes



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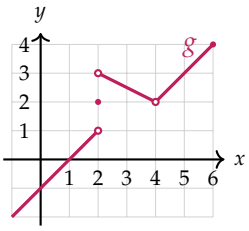
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TALK · 2 minutes



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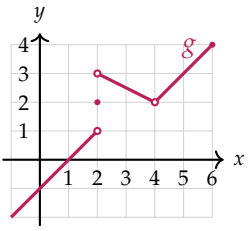
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sec

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VOTE AGAIN



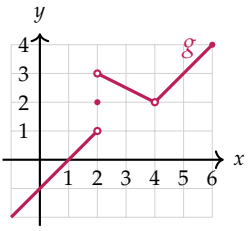
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Eyes on the screen. Hands up when your letter appears — hold for three seconds.

A B C D

VOTE AGAIN



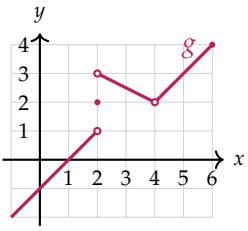
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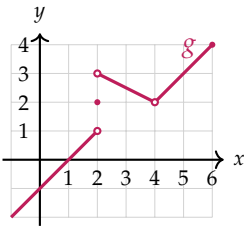
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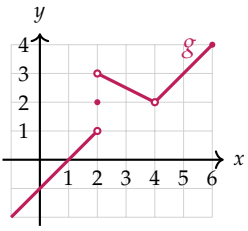
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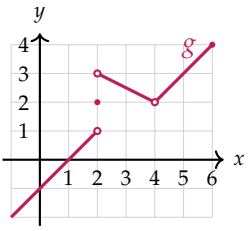
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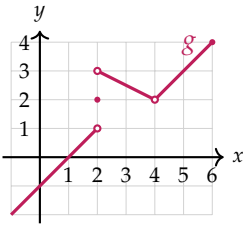
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Hands down. Thank you.

A B C D

MIC

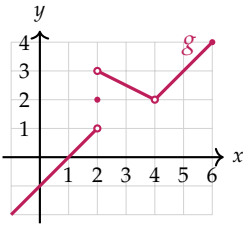


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The microphone goes to

the team in the middle block, row 7, next to the right aisle — runner, please. Give the reason; I repeat what you say.



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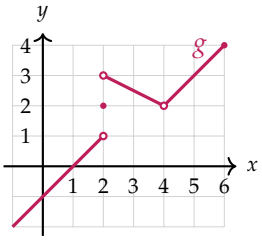
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Answer

C. Left: $g(x) \to 1$; right: $g(x) \to 3$. Different one-sided limits, so no limit. D is the value $g(2) = 2$, not the limit; A and B are one side each.

STAND UP

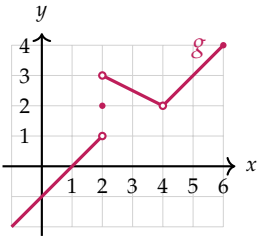


"$\lim_{x \to 4} g(x)$ exists."

Stand up if you think this is TRUE.

Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

STAND UP



"$\lim_{x \to 4} g(x)$ exists."

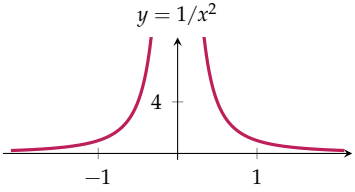
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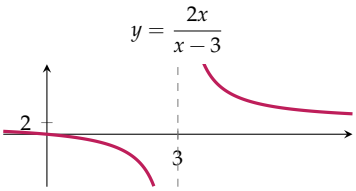
Stay standing until I say "sit". Then find someone with the other opinion — 60 seconds.

Verdict

TRUE. From both sides $g(x) \to 2$, so $\lim_{x \to 4} g(x) = 2$ — although $g(4)$ is undefined (the hole). A limit does not need the value.

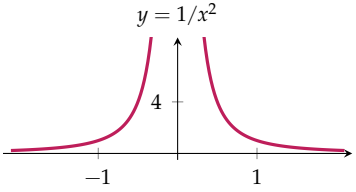
Infinite limits and vertical asymptotes

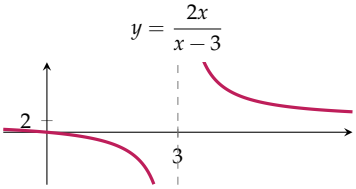




$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

Infinite limits and vertical asymptotes

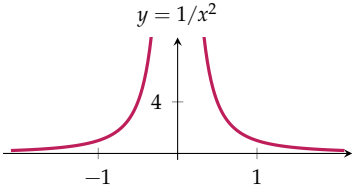




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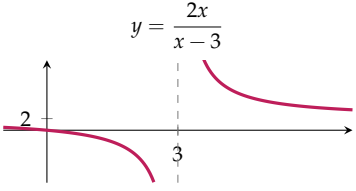
This does not say the limit is a number called ∞. It says: $1/x^2$ goes through the roof — as large as we like — when x is close enough to 0.

Infinite limits and vertical asymptotes



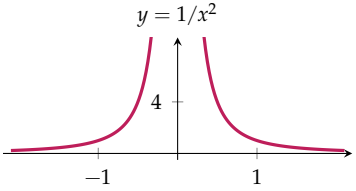
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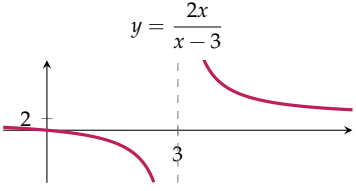
$$\lim_{x \to 3^+} \frac{2x}{x-3} = \infty, \qquad \lim_{x \to 3^-} \frac{2x}{x-3} = -\infty$$

Infinite limits and vertical asymptotes



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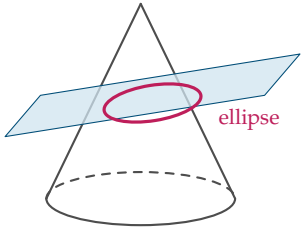


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$x = a$ is a vertical asymptote of $y = f(x)$ if at least one one-sided limit at a is $\pm\infty$. Here: $x = 3$.

A little history · 1655

The symbol ∞ was introduced by John Wallis in 1655, in a book on conic sections — the curves you get by cutting a cone with a plane: an ellipse, a parabola, a hyperbola. Nobody knows for certain why he chose the sign; a favourite guess is the Roman numeral for 1000, CIO, written as one stroke. Wallis, too, used it for "as large as we like", not for a number.



STAND UP

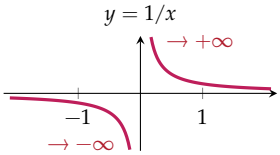
$$\text{“}\lim_{x\to 0}\frac{1}{x}=\infty\text{”}$$

Stand up if you think this is TRUE.

Stay standing until I say “sit”. Then find someone with the other opinion — 60 seconds.

STAND UP

$$\text{“}\lim_{x\to 0}\frac{1}{x}=\infty\text{”}$$



Stand up if you think this is TRUE.

Stay standing until I say “sit”. Then find someone with the other opinion — 60 seconds.

Verdict

FALSE. For $x < 0$ the values $1/x$ are large *negative*: $\lim_{x\to 0^-} 1/x = -\infty$ while $\lim_{x\to 0^+} 1/x = +\infty$. The two sides disagree, so we cannot even write ∞ for the two-sided limit. Compare: $\lim_{x\to 0} 1/x^2 = \infty$ *is* correct, because both sides go up.

Today in three lines

1. $\lim_{x \to a} f(x) = L$ is about x near a, never at a: the value $f(a)$ may be L, undefined, or something else.

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One of today's questions will be on the quiz

Reread Questions 1 and 2 tonight. Then Stewart §1.5: Examples 1–9, Exercises 4–11 (read the graph), 29–37 (infinite limits).

EXIT TICKET

Tear a quarter page from a notebook — one per team.

1. Team name (and number, once you have one); names of those present today.
2. One sentence: *"The limit of f at a is …" — and hand in your team registration page with the ticket.*
3. Your muddiest point — the one thing still unclear.
4. One moment today that surprised you (or made you smile).

Pass it to the aisle; runners collect. Thank you.

BREAK
10 minutes

Stand up, stretch, then sit down again with your team.

Next: calculating limits with the limit laws (Lecture 2).



Course channel on Telegram:
For Every Epsilon
@forever_epsilon